

Predicting Performance in NYC High Schools

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Introduction and Dataset

For this series of projects, I focused my analysis on the NYC High School Performance Dataset, which I sourced directly from New York City Department of Education (DOE). The dataset provides a granular look at the educational metrics for 410 unique public high schools across New York City between the years 2021-2023. While the original raw file contained a wide array of variables, I narrowed the scope down to 15 key features that specifically capture school staffing metrics, teacher demographics, and student demographics. This selection was intentional; I wanted to isolate the internal "inputs" of the school system, who works there and who attends, to see if they could predict outcomes without any knowledge of the curriculum or funding levels.

This project is deeply connected to my ongoing Capstone project, which investigates how external city systems like zoning, land use, and public transit influence student success. While my Capstone looks outward at the neighborhood, this project looked inward at the institution. I chose this specific dataset because it allowed me to test a fundamental hypothesis: that the structural characteristics of a school are strong predictors of its academic output. Specifically, I wanted to determine if we could predict a school's performance category solely based on staffing ratios and demographic composition.

The target variable for my prediction models was "Performance," a categorical variable I derived from the schools' mean Regents exam scores. I classified the schools into three balanced classes: Low, Medium, and High performance. This transformation turned a regression problem into a supervised classification task, allowing me to see if machine learning algorithms could distinguish between struggling, average, and high-achieving schools based purely on their structural data. My central hypothesis going into this project was that staffing ratios, specifically the number of students per teacher, would hold much stronger predictive power than raw staffing counts. My reasoning was that a ratio reflects an institution's actual capacity to serve its students, whereas raw counts often just reflect the sheer size of the building.

Data Cleaning and Representation

The reality of working with public data is that it is rarely ready for immediate analysis, and this dataset was no exception. The initial cleaning process required a close inspection of missing values to ensure the models wouldn't be biased by gaps in the record. While most of the staffing metrics were complete, I identified missing data in the `Male_Teacher_Percentage` and

Female_Teacher_Percentage columns, affecting approximately 4.3% of the training set. To handle this, I had to choose between dropping those rows or filling them in. I opted to use `SimpleImputer(strategy='median')`. I chose the median over the mean because the staffing data was highly skewed; a few massive schools with unusual demographic profiles could have pulled the mean away from the "typical" school profile, whereas the median provided a more robust measure of central tendency.

Once the gaps were filled, I had to address the representation of the data. I split the dataset into training and testing sets, keeping 80% for training and reserving 20% for final validation. A crucial step here was using a stratified split. Because I was dealing with three distinct performance classes (Low, Medium, High), I needed to ensure that the proportions of these classes were identical in both the training and testing sets. Without stratification, a random split could have accidentally stacked the training set with "High" performing schools, teaching the model to guess "High" every time.

The most significant challenge I faced during the data exploration phase was handling the massive variance in school sizes. New York City's school system includes everything from massive comprehensive high schools with thousands of students to tiny, specialized academies. Raw integers for "Teacher Count" or "Total Staff" were acting as proxies for school size rather than school quality. To fix this, I created new features focused on ratios, such as `Student_Teacher_Ratio`. This normalization was critical because it allowed the algorithms to compare a giant school in Queens with a small school in the Bronx on equal footing, focusing on their instructional capacity rather than their square footage.

Visualization

Before running any models, I spent time visualizing the data, which revealed deep structural inequalities within the system. When I plotted histograms of the staffing metrics, student-teacher ratios ranged from as low as 0.74 to as high as 19.16. This massive range indicated that before we even talk about performance, there is a fundamental disparity in instructional capacity across the city.

Segmenting these visualizations by performance category confirmed my initial hypothesis. When I broke the data down, high-performing schools exhibited a mean student-teacher ratio of 3.07, whereas low-performing schools averaged 3.89. The histograms clarified that while high-performing schools often employed more staff overall, their true advantage lay in maintaining significantly more favorable ratios. This was a key insight: quality in this dataset looks like "more teachers per student," not just "more teachers."

However, the visualization phase also gave me a warning about the difficulty of the classification task ahead. Scatter matrices showed that while trends existed, the data did not separate into clean, distinct clusters. The "Medium" performance schools overlapped significantly with both the High and Low categories. There was no clear boundary line where a school stopped being "Low" and started being "Medium." This visual evidence suggested that

while resources like staffing are informative, the relationship is messy and non-linear, foreshadowing that simple models might struggle to accurately classify the schools stuck in the middle.

Supervised Learning Experiments

For the supervised learning portion of the project, I experimented with two distinct algorithms to see which could better handle this structural complexity: K-Nearest Neighbors (KNN) and Random Forest.

I started with K-Nearest Neighbors. In our case, if a school's staffing numbers look geometrically like five other schools that are "High Performing," the algorithm assumes this new school is also "High Performing." I tuned the hyperparameter k , using GridSearchCV, testing values from 1 to 25. The results were disappointing. The default model with $k=5$ achieved a testing accuracy of roughly 46%, which is barely better than random guessing. The tuning process suggested that $k=1$ was the optimal parameter, but this was a false positive that led to severe overfitting; it memorized the training data perfectly but failed to generalize to new schools, hitting only 57% accuracy on the test set.

The failure of KNN was instructive. KNN relies on geometric distance, assuming that schools with similar staffing numbers are "close" to each other. However, in the NYC school system, two schools can have identical staffing ratios but vastly different outcomes due to unmeasured factors like school culture, funding, or neighborhood safety. "Distance" in this dataset was not a good proxy for similarity in performance, causing the algorithm to fail.

I then moved to Random Forest, which uses a completely different logic. Each tree asks a series of yes/no questions about the data (e.g., "Is the student-teacher ratio lower than 3.5?"), and the algorithm averages the predictions of all these trees to make a final decision. I tuned three key parameters: the number of trees ($n_estimators$), the maximum depth of the trees (max_depth), and the minimum samples required to split a node.

Random Forest proved to be the superior model for this task. After tuning the model to use 500 trees, it achieved a testing accuracy of roughly 60%. While this isn't perfect, it was significantly better than KNN. The reason for this improvement became clear during the feature importance analysis. Random Forest successfully identified that `Student_Teacher_Ratio` was the single most important predictor. Unlike KNN, which got confused by distance, the Random Forest was able to identify specific threshold effects, capturing the non-linear reality that once staffing ratios cross a certain critical point, performance tends to drop.

PCA for Feature Selection

After establishing a baseline with the supervised models, I wanted to see if reducing the complexity of the data could improve the results. I applied Principal Component Analysis (PCA), a technique that compresses the information in the dataset into fewer, abstract dimensions. I

determined that I could reduce the dataset from 15 original features down to just 9 principal components while still retaining 95% of the variance.

I retrained the best-performing Random Forest model using these 9 abstract components instead of the original 15 features. The result was a decrease in performance. The accuracy dropped from approximately 65% with the original features down to 57% with the PCA components. This negative result was fascinating because it highlighted how Random Forest works. The algorithm thrives on making decision cuts on specific, interpretable features, like splitting schools based on their exact teacher count. When I transformed those crisp features into abstract mathematical components (e.g., "Principal Component 1"), I removed the direct signals the decision trees relied on. In this specific context, simplifying the data actually made the problem harder for the model to solve.

Unsupervised Learning and Clustering

In the final phase, I shifted to unsupervised learning. Here, I stripped away the "Performance" labels entirely and asked the algorithms to group the schools based solely on their structural similarities. I used PCA as a pre-processing step for three algorithms: K-Means, Hierarchical Clustering, and DBSCAN. I then compared the clusters these algorithms found against the actual performance labels using the Adjusted Rand Index (ARI).

The K-Means algorithm attempts to partition the schools into k distinct groups by minimizing the distance between data points and the center of their group. I set $k=3$ to match our known performance classes. The results were stark: the ARI score hovered near 0.01, meaning the clusters had almost zero correlation with the actual performance tiers. Whether I used the original data or the PCA-reduced data, the algorithm found stable groups of schools, but those groups did not align with "High," "Medium," or "Low" performance.

Hierarchical Clustering, which builds a tree of relationships between schools, yielded similarly poor results, with ARI scores dropping even further when PCA was applied. Finally, DBSCAN, a density-based algorithm, largely failed on this dataset. It categorized the vast majority of schools as "noise" when PCA was applied, unable to find dense regions of similar schools in the data.

These failures are a major finding. They illustrate the "ecological fallacy", the idea that group-level characteristics (like staffing) can perfectly predict outcomes. The unsupervised analysis proved that schools with identical structures can have completely different academic results. There is no distinct "shape" of a high-performing school in the data; high performance exists across the spectrum of school sizes and staffing profiles.

Summary and Reflection

The most significant finding across these three projects is that machine learning models are constrained by the scope of their input data. While often viewed as tools for uncovering hidden

patterns, these algorithms function primarily as quantifiers of the information provided. In this analysis, the results highlighted a fundamental structural reality of the NYC public school system. The Random Forest model validated the core hypothesis: resource allocation, specifically staffing ratios, is a stronger predictor of performance than the raw scale of a school. The feature importance analysis confirmed that instructional capacity is a primary driver of student outcomes, providing statistical evidence that smaller class sizes and better ratios correlate with higher performance. This finding aligns with the sociological theories described by Anthony Bryk, suggesting that the "communal organization" of a school, measured here through staffing availability, is essential for effective education.

The limitations of the analysis were equally instructive. The unsupervised clustering algorithms failed to separate schools into distinct performance categories based on internal metrics. This is a significant negative result: "High Performing Schools" do not exist as a distinct structural type in this dataset. A high-performing school in one borough often shares the exact same staffing profile as a struggling school in another. This inability to distinguish schools based on internal metrics alone indicates that the "Performance" variable is driven by factors external to the dataset. While adequate staffing is necessary for success, the analysis demonstrates it is not sufficient to explain the variance in outcomes. The prediction accuracy capped at roughly 67% because the model lacked data on the socioeconomic context of the student body and the neighborhood, factors like poverty and housing stability that internal school restructuring cannot address.

This conclusion has necessitated a shift in strategy for my Capstone project. Modeling schools as independent entities in a machine learning classifier is a methodological error because it ignores the environmental context. To address this, my Capstone will move beyond "black box" prediction and emulate the statistical framework used by Anthony Bryk, Valerie Lee, and Peter Holland in *Catholic Schools and the Common Good*. Their research demonstrated that accurate models of school effectiveness must account for the social environment.

Following their approach, I will replace standard classification with Hierarchical Linear Modeling (HLM). By nesting schools within their specific neighborhoods, this framework will allow me to isolate the internal effects of school organization from the external pressure of city systems. This method mathematically respects the hierarchical nature of the data: students function within schools, and schools function within communities. This approach will allow me to quantify the extent to which student success is attributable to internal factors (like staffing ratios) versus external policy contexts (like zoning density and transit access). This shifts the analytical focus from simply predicting which schools succeed to explaining the structural mechanisms that distribute success so unevenly. Ultimately, this project served as a proof of concept, demonstrating that accurate modeling of school performance requires measuring the neighborhood structure that surrounds the institution.